

- (c) Many electrons are now accelerated from rest from plate X to plate Y in Fig. 6.1. When the electrons hit plate Y, the absorption of their kinetic energies results in the emission of electromagnetic waves.

(i) Show that the minimum wavelength of these electromagnetic waves is 21 pm.

[3]

- (ii) State the region of the electromagnetic spectrum that contains these waves.

..... [1]

- (iii) Explain how these electromagnetic waves may be used to form images of internal body structures.

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..... [2]

[Total: 11]

